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METHODS AND SYSTEMS FOR SECURING BEARING ELEMENTS

Abstract

Methods of securing bearing elements to bearing rings are disclosed. The methods include clamping the bearing elements in place on the bearing housing and applying heat to braze the bearing elements in place while clamped. The heat may be applied using a flame ring.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] The present application claims the benefit of U.S. Provisional Patent Application No. 63/562,080 (pending), filed on Mar. 6, 2024, and entitled “Methods of Securing Bearing Elements,” the entirety of which is incorporated herein by reference and made a part of the present disclosure.

FIELD
[0002] The present disclosure relates to methods and systems for securing bearing elements, to bearings made with such methods and systems, and to systems including such bearings.
BACKGROUND
[0003] Bearing elements are sometimes secured to a bearing ring via brazing the bearing element in place on the bearing ring. Typically, after the brazing is performed, it is necessary to perform post-brazing machining in order to more precisely position the bearing elements relative to the bearing ring and/or ensure proper tolerances in the bearing assembly. It would be desirable to have a method of securing bearing elements that more precisely positions bearing elements on a bearing ring without requiring post-brazing processing.
BRIEF SUMMARY
[0004] Some embodiments of the present disclosure include a method of securing bearing elements to a bearing housing. The method includes positioning a plurality of bearing elements within sockets of a bearing housing, and positioning a braze material relative to the bearing elements and the bearing housing. The method includes applying a clamping force to the bearing elements to force the bearing elements into the sockets. While applying the clamping force, the method includes applying heat to the bearing elements, the braze material, and the bearing housing. The heat initiates melting of the braze material to braze the bearing elements into the sockets.
[0005] Some embodiments of the present disclosure include a bearing assembly made by the methods disclosed herein.
[0006] Some embodiments of the present disclosure include a clamp assembly for applying a clamping force to bearing elements during brazing. The clamp assembly includes a connecting rod having a first end and a second end. A first clamp is coupled with the first end of the connecting rod, and a second clamp is coupled with the second end of the connecting rod. A spring is coupled with the second end of the connecting rod such that the second clamp is positioned between the spring and the first clamp, Compression of the spring is adjustable.
[0007] Some embodiments of the present disclosure include a clamped bearing assembly for brazing. The assembly includes a bearing housing having a first end with a first bearing ring and a second end with a second bearing ring. The assembly includes a connecting rod having a first end and a second end. The connecting rod extends through the bearing housing such that the first end of the connecting rod extends from the first end of the bearing housing and the second end of the connecting rod extends from the second end of the bearing housing. A first clamp is coupled with the first end of the connecting rod, and a second clamp is coupled with the second end of the connecting rod. The first and second bearing rings are positioned between the first and second clamps. A spring is coupled with the second end of the connecting rod such that the second clamp is positioned between the spring and the second bearing ring, Compression of the spring is adjustable to apply a compression force onto the first and second bearing rings with the first and second clamps.
[0008] Some embodiments of the present disclosure include a method of securing bearing elements to a bearing ring. The method includes positioning a plurality of bearing elements on a bearing ring, positioning a braze material on the bearing elements, the bearing body, or combinations thereof, and applying a force to the bearing elements, forcing the bearing elements toward the

bearing body. While applying the force, the method includes applying heat from one or more flames of a burner to the bearing elements, the braze material, the bearing body, or combinations thereof. The heat at least partially melts the braze material, brazing the bearing elements onto the bearing ring.

[0009] Some embodiments of the present disclosure include a clamp assembly for applying a clamping force to bearing elements during brazing of the bearing elements to a bearing ring. The clamp assembly includes a base plate defining a surface configured to receive a bearing ring with bearing elements. The clamp assembly includes a braze cap having an engagement surface. A spring is positioned on the braze cap such that the braze cap is positioned between the spring and the base plate. An actuator is positioned to compress the spring to apply a force to the braze cap to force the braze cap toward the base plate.

[0010] Some embodiments of the present disclosure include a flame ring for applying heat to braze bearing elements to a bearing ring. The flame ring includes a gas manifold defining a central annulus. A plurality of nozzles are in fluid communication with the gas manifold. The nozzles are positioned to direct flame from the nozzles toward the central annulus.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0011] So that the manner in which the features and advantages of the assemblies, systems, and methods of the present disclosure may be understood in more detail, a more particular description briefly summarized above may be had by reference to the embodiments thereof which are illustrated in the appended drawings that form a part of this specification. It is to be noted, however, that the drawings illustrate only various exemplary embodiments and are therefore not to be considered limiting of the disclosed concepts as it may include other effective embodiments as well.

[0012] FIG. **1**A is a top view of an assembly of a thrust bearing ring clamped during brazing in accordance with embodiments of the present disclosure.

[0013] FIG. **1**B is a cross-sectional view of the assembly of FIG. **1**A along line A-A in accordance with embodiments of the present disclosure.

[0014] FIG. **1**C is a detail view of a portion of FIG. **1**B showing engagement between portions of a clamp and a thrust bearing element in accordance with embodiments of the present disclosure.

[0015] FIG. **1**D is a slice view of the assembly of FIG. **1**A along line A-A in accordance with embodiments of the present disclosure.

[0016] FIG. **1**E is a side view of the assembly of FIG. **1**A in accordance with embodiments of the present disclosure.

[0017] FIG. **1**F is a perspective view of the assembly of FIG. **1**A in accordance with embodiments of the present disclosure.

[0018] FIG. **1**G is an alternative cross-sectional view of the assembly of FIG. **1**A along line A-A but with the addition of a heat shield positioned around a spring of the clamp in accordance with embodiments of the present disclosure.

[0019] FIG. **2**A is a top view of an assembly of a radial bearing ring clamped during brazing in accordance with embodiments of the present disclosure.

[0020] FIG. **2**B is a cross-sectional view of the assembly of FIG. **2**A along line B-BA in accordance with embodiments of the present disclosure.

[0021] FIG. **2**C is a detail view of a portion of FIG. **2**B showing engagement between portions of a clamp and a radial bearing element.

[0022] FIG. **2**D is a slice view of the assembly of FIG. **2**A along line A-A in accordance with embodiments of the present disclosure.

[0023] FIG. **2**E is a side view of the assembly of FIG. **2**A in accordance with embodiments of the present disclosure.
[0024] FIG. **2**F is a perspective view of the assembly of FIG. **2**A in accordance with embodiments of the present disclosure.
[0025] FIG. **3**A is a top view of an assembly of an angular bearing ring clamped during brazing in accordance with embodiments of the present disclosure.
[0026] FIG. **3**B is a cross-sectional view of the assembly of FIG. **3**A along line C-C in accordance with embodiments of the present disclosure.
[0027] FIG. **3**C is a detail view of a portion of FIG. **3**B showing engagement between portions of a clamp and an angular bearing element.
[0028] FIG. **3**D is a slice view of the assembly of FIG. **3**A along line C-C in accordance with embodiments of the present disclosure.
[0029] FIG. **3**E is a side view of the assembly of FIG. **3**A in accordance with embodiments of the present disclosure.
[0030] FIG. **3**F is a perspective view of the assembly of FIG. **3**A in accordance with embodiments of the present disclosure.
[0031] FIG. **4** is a schematic of a flame ring coupled with a gas/air supply in accordance with embodiments of the present disclosure.

DETAILED DESCRIPTION

[0032] Embodiments of the present disclosure include methods and systems for securing bearing elements to bearing assemblies. The bearing elements may be thrust bearing elements, radial bearing elements, or angular bearing elements. The bearing assemblies may be or include bearing rings. The bearing elements can be secured to the bearing rings via brazing, including the application of heat to a braze material positioned adjacent the bearing elements and bearing assemblies. The heat can be provided by a flame, induction, or another heat source.

Brazing Thrust Bearings

[0033] With reference to FIGS. **1**A-**1**G, a method and system for securing thrust bearing elements to a thrust bearing ring is described. Brazing system **1000** is configured to secure a bearing ring for brazing of bearing elements thereto. In FIGS. **1**A-**1**G, the brazing system **1000** is depicted brazing thrust bearing elements **106** to a thrust bearing ring **102**. However, the brazing systems disclosed herein are not limited to use on thrust bearings and may be used on other bearings, such as radial bearings and angular bearings.
[0034] Brazing system **1000** includes a clamp assembly **122** and a flame ring **114**. The clamp assembly **122** is configured to secure a position of the thrust bearing ring **102** and thrust bearing elements **106** and to apply a compressive force to the thrust bearing elements **106** during the brazing process. The flame ring **114** is configured to provide flames to heat the thrust bearing ring **102**, thrust bearing elements **106**, and braze material **108** to achieve the brazing of the thrust bearing elements **106** onto the thrust bearing ring **102**.
[0035] The clamp assembly **122** includes a chuck **100** having a chuck jaw **118**. A base plate **124** of the clamp assembly **122** is positioned and secured in the chuck jaws **118**. Base plate **124** includes a threaded hole **120**. The clamp assembly **122** includes a braze cap **110** having a clearance hole **126**. To secure the thrust bearing ring **102** in the clamp assembly **122**, the thrust bearing ring **102** is positioned onto the base plate **124**. The thrust bearing ring **102** includes sockets **104** (or other receptacles) configured (e.g., shaped and sized) to receive the thrust bearing elements **106**. The braze material **108** is positioned within the sockets **104** on the thrust bearing ring **102** along with an appropriate flux. The thrust bearing elements **106** are positioned within the sockets **104** such that the braze material **108** is positioned at least partially between the thrust bearing ring **102** and the thrust bearing elements **106**. The thrust bearing elements **106** can be polycrystalline diamond bearing elements, such as a polycrystalline diamond compact (PDC). The braze material **108** can be a braze alloy or solder alloy in the form of a disc, paste, wire ring, or other form.

[0036] With the thrust bearing ring **102** and thrust bearing elements **106** in place, the braze cap **110** is secured over the thrust bearing ring **102**. The braze cap **110** is positioned over the thrust bearing elements **106** such that the thrust bearing elements **106** are positioned between the braze cap **110** and the thrust bearing ring **102**. The braze material **108** is positioned between a bottom surface **105** of the thrust bearing element **106** and a bottom surface **103** of the socket **104**. A bottom surface **111** of the braze cap **110** is engaged with a top surface **107** of the thrust bearing element **106**. Thus, the braze cap **110** is arranged to force the thrust bearing element **106** into the socket **104**.
[0037] With the braze cap **110** in place, the braze cap **110** is secured in position via a bolt **112**, spring **128**, and washer **130**. The bolt **112** is engaged through the spring **128**, washer **130**, clearance hole **126**, and thrust bearing ring **102**. The bolt **112** is threaded with the base plate **124** at threaded hole **120**.
[0038] The clamp assembly **122** secures the thrust bearing elements **106** to the thrust bearing ring **102** during the brazing process to facilitate accurate and precise positioning of the thrust bearing elements **106** on the thrust bearing ring **102**. The clamp assembly **122** exerts a force on the thrust bearing elements **106**, pressing the thrust bearing elements **106** toward the thrust bearing ring **102** to maintain the thrust bearing elements **106** in position within the socket **104** during the brazing process. The direction of the force exerted onto the bearing elements **106** via the clamp assembly **122** can vary depending on the particular application and type of bearing element. In the embodiment shown in FIGS. **1**A-**1**G, the force is in direction **101**. While a clamp assembly is shown and described in the present disclosure, the methods and systems disclosed herein are not limited to use of a clamp assembly and may include other structures configured to force the bearing elements into a desired position on the bearing ring.
[0039] The bolt **112** can be tightened or loosened to increase or decrease compression forces exerted onto the spring **128**, and in-turn onto the braze cap **110** and thrust bearing elements **106**. Thus, the force (clamping force) applied onto the thrust bearing elements **106** during brazing can be controlled and adjusted. The clamping pressure applied to the seating face of the thrust bearing elements **106** by the clamp assembly **122** to the thrust bearing elements **106** can range from greater than 0 psi to 1,000 psi; or from 50 psi to 950 psi; or from 100 psi to 900 psi; or from 150 psi to 850 psi; or from 200 psi to 800 psi; or from 250 psi to 750 psi; or from 300 psi to 700 psi; or from 350 psi to 650 psi; or from 400 psi to 600 psi; or from 450 psi to 550 psi; or any range or discrete value therebetween. The spring **128** used in the clamp assembly **122** can be varied to vary the maximum amount of force available to be applied during the brazing process. Some exemplary springs suitable for use in certain embodiments include a spring having a maximum clamping force of 424 lbs., a spring having a maximum clamping force of 618 lbs., a spring having a maximum clamping force of 990 lbs., and a spring having a maximum clamping force of 1,634 lbs. In some embodiments, the maximum clamping force of the spring is from 300 lbs. to 2,000 lbs, or from 400 lbs. to 1,900 lbs., or from 500 lbs. to 1,800 lbs., or from 600 lbs. to 1,700 lbs., or from 700 lbs. to 1,600 lbs., or from 800 lbs. to 1,500 lbs., or from 900 lbs. to 1,400 lbs., or from 1,000 lbs. to 1,300 lbs., or from 1,100 lbs. to 1,200 lbs., or any range or discrete value therebetween. The springs are not limited to these exemplary properties and may have a maximum clamping force less than 300 lbs. or more than 2,000 lbs.
[0040] While clamping and use of a clamping assembly is described herein for applying a seating force to the bearing elements before and during the brazing, the methods disclosed herein are not limited to use of clamping and may include other methods of proving a seating force to the bearing elements. For example, the seating force can be applied to the bearing elements using weight, hydraulic force, or pneumatic force. In one embodiment, a first end of the bearing housing is placed on a surface and weight is placed on a second, opposite end such that the weight (e.g., via gravity) exerts a force on the second end and forces the first end to press against the surface, with the surface and the weight applying the seating forces to the bearing elements at the ends of the bearing housing. In another embodiment, a first end of the bearing housing is placed on a surface and

hydraulic piston is coupled between the surface and a plate on the second, opposite end of the bearing housing. The hydraulic piston can actuate to move the plate towards the surface with the bearing housing positioned between the plate and the surface such that the plate exerts a force on the second end and forces the first end to press against the surface, with the surface and the plate applying the seating forces to the bearing elements at the ends of the bearing housing.

[0041] The spring **128** can be made of an alloy capable of withstanding the high-temperatures of the brazing process without melting or other structural failure occurring. For example, the spring **128** can be Inconel alloy or steel. As shown in FIG. **1**G, in some embodiments, a heat shield **129** is positioned around the spring **128**, such that the spring **128** is positioned between the heat shield **129** and the bolt **112**. The heat shield **129** can protect the spring during brazing, preventing or reducing the flame of the brazing process from affecting the mechanical and structural integrity of the spring **128**. The heat shield **129** can include a ceramic fiber material (e.g., FIBERFRAX®) wrapped around the spring **128** to help thermally insulate the spring **128** so that the flame does not heat the spring **128** sufficiently to affecting the mechanical and structural integrity of the spring **128**.

[0042] With the thrust bearing ring **102** and thrust bearing elements **106** secured in place by the clamp assembly **122** at the desired compressive force, the brazing of each of the thrust bearing elements **106** can be conducted using the flame ring **114**.

[0043] The flame ring **114** includes a gas manifold **132**. In the depicted embodiments, the gas manifold **132** is in the shape of a ring or circle. However, the flame rings disclosed herein are not limited to having such a shape. Furthermore, while referred to as a “ring” the flame rings disclosed here may be other shapes. The flame ring **114** includes a gas inlet **134** in fluid communication with the gas manifold **132** to provide flammable gas, from a gas source (not shown), into the gas manifold **132**. The flame ring **114** includes nozzle fittings **136** in fluid communication with the gas manifold **132** and configured to receive flammable gas therefrom. The flame ring **114** includes nozzles **138** are in fluid communication with the nozzle fittings **136** to receive flammable gas therefrom. The nozzles **138** can be connected with the nozzle fittings **136** via quick connects and fish tail tips connected with the nozzles **138**. The nozzles **138** are posited to expel flammable gas toward the thrust bearing ring **102**, thrust bearing elements **106**, and braze material **108**. While not depicted, the flame ring **114** may include an ignition system (e.g., spark plug) positioned to ignite the flammable gas exiting the nozzles **138** to form flames **116** or can be ignited manually with an external ignition source. The nozzles **138** can include braze tips positioned to direct the flames **116** toward a portion of thrust bearing ring **102**.

[0044] The flames **116** are directed toward and/or onto the thrust bearing elements **106**, braze material **108**, and thrust bearing ring **102** to at least partially melt the braze material **108**. The melting of the braze material **108** results in the brazing of the thrust bearing elements **106** to the thrust bearing ring **102**. During the melting of the braze material **108**, the spring **128** exerts a compressive force on the thrust bearing elements **106** to force the thrust bearing elements **106** to seat within the sockets **104** of the thrust bearing ring **102**.

[0045] The flame ring **114** is positioned about and encircles or otherwise surrounds the thrust bearing ring **102** and includes a plurality of the nozzles **138** to substantially surround the thrust bearing ring **102** with the flames **116**. The flame ring **114** and plurality of nozzles **138** can be positioned relative to the thrust bearing ring **102** such that when the flame ring **114** produces the flames **116**, the flames **116** are directed to heat the thrust bearing ring **102**, braze material **108**, and plurality of thrust bearing elements **106** to simultaneously braze the plurality of thrust bearing elements **106** to the thrust bearing ring **102**. Various aspects of the flame ring **114** can be varied to vary the amount of heat applied to the thrust bearing elements **106** such as the angles of the nozzles **138** (burners). For example, the angles of the nozzles **138** can be adjustable such that the various flames **116** impinge upon the thrust bearing ring **102** at different angles. Also, the number of nozzles **138**, distance that the flames **116** extends from the nozzles **138**, and gas mixture that the flame ring **114** burns can be varied. The diameter of the gas manifold **132** can be varied to accommodate for use with various sized bearing rings.

[0046] In some embodiments, during the application of the flames **116** to the thrust bearing ring **102**, the thrust bearing ring **102** is rotated while the flame ring **114** is held static. With reference to FIG. **1**A, the thrust bearing ring **102** can be rotated in rotational direction **140** during the brazing. Rotating the thrust bearing ring **102** can provide for a more even heating of the thrust bearing elements **106**. The thrust bearing ring **102** can be rotated at from greater than 0 rpm to 1,000 rpm, from 50 to 950 rpm, from 100 to 900 rpm, from 150 to 850 rpm, from 200 to 800 rpm, from 250 to 750 rpm, from 300 to 700 rpm, from 350 to 650 rpm, from 400 to 600 rpm, from 450 to 550 rpm, or any range or discrete value therebetween. The rpm of the thrust bearing ring **102** can be constant or can vary over the duration of the brazing process. While the thrust bearing ring **102** is described as being rotated, with the flame ring **114** being held static, in other embodiments the flame ring **114** rotates and the thrust bearing ring **102** is held static. In still other embodiments, both the flame ring **114** and the thrust bearing ring **102** are rotated. The flames **116** may touch the thrust bearing ring **102** or may be spaced-apart from the thrust bearing ring **102** while still providing sufficient heat to melt the brazing material **108**.

[0047] In some embodiments, the flames **116** form a reducing atmosphere in proximity to the thrust bearing ring **102**, providing for less oxidation of the components of the thrust bearing ring **102**. In some embodiments, the brazing is performed under vacuum (e.g., in a vacuum furnace) or in an inert gas atmosphere.

[0048] In some embodiments a temperature sensor (not shown), such as a thermocouple or IR sensor, can be used to monitor the temperature of the thrust bearing ring **102** and/or thrust bearing elements **106** during the brazing process. In some embodiments, temperature-indicating crayons are used to monitor the temperature of the thrust bearing ring **102** and/or thrust bearing elements **106** during the brazing process.

[0049] In certain embodiments, the brazing process of the thrust bearing elements **106** to the thrust bearing ring **102** can be performed in less than 10 minutes, or in 3 to 5 minutes.

[0050] Once the brazing of the thrust bearing elements **106** is completed, the clamping assembly can be removed by unthreading the bolt **112**, and the thrust bearing ring **102** can be removed from the brazing system **1000**. In some embodiments, after the thrust bearing elements **106** are brazed to the thrust bearing ring **102**, no post-processing is performed on the brazed bearing elements **106**.

[0051] The application of the clamping force by the clamping assembly **122** during the brazing provides the ability to enhance the accuracy and precision of the positioning of the thrust bearing elements **106** in the thrust bearing ring **102**. For example, each thrust bearing element **106** on the thrust bearing ring **102**, after brazing, can extend above a surface of the thrust bearing ring **102** to the same or substantially the same height.

Brazing Radial Bearings

[0052] With reference to FIGS. **2**A-**2**F, a method and system for securing radial bearing elements to a radial bearing ring is described. Brazing system **2000** is configured to secure a thrust bearing ring for brazing of the thrust bearing elements thereto. In FIGS. **2**A-**2**F, the brazing system **2000** is depicted brazing radial bearing elements **206** to a radial bearing ring **202**.

[0053] Brazing system **2000** includes a clamp assembly **222** and a flame ring **214**. The clamp assembly **222** is similar to the clamp assembly **122** shown in FIGS. **1**A-**1**G, but includes different components that are arranged to secure a position of the radial bearing ring **202** and radial bearing elements **206** and to apply a compressive force to the bearing elements **206** during the brazing process. The flame ring **214** is substantially similar, and may be identical in embodiments, to the flame ring **114** shown in FIGS. **1**A-**1**G. Flame ring **214** is configured to provide flames to heat the radial bearing ring **202**, radial bearing elements **206**, and braze material **208** to achieve the brazing of the radial bearing elements **206** onto the radial bearing ring **202**.

[0054] The clamp assembly **222** includes a chuck **200** having a chuck jaw **218**. A base plate **224** of the clamp assembly **222** is positioned and secured in the chuck jaws **218**. Base plate **224** includes a threaded hole **220**. The clamp assembly **222** is different from the clamp assembly **122** at least in that rather than including only one braze cap **110**, the clamp assembly **222** includes two braze caps including a first braze cap **210***a* having a first clearance hole **226***a* and a second braze cap **210***b* having a second clearance hole **226***b*. Each of the first and second braze caps **210***a* and **210***b* include an angled surface **213**. For example, and without limitation, the angled surfaces **213** may be beveled surfaces, chamfered edges, or another transitional surface between the surface **211** and the side edge **215** of the braze caps **210***a* and **210***b*. A concavity **217** is defined by the angled surfaces **213** between the braze caps **210***a* and **210***b*. The clamp assembly **222** includes a radial force member **221** positioned between the radial bearing elements **206** and the braze caps **210***a* and **210***b*. The radial force member **221** is positioned at least partially in the cavity **217** and in engagement with the angled surfaces **213**. The radial force member **221** is configured to translate axial forces on the braze caps **210***a* and **210***b* into radial forces directed toward the radial bearing elements **206**. Prior to compression, the braze cap **210***a* can be spaced apart from the braze cap **210***b*. As the braze cap **210***a* is forced downward toward the braze cap **210***b* via force from the bolt **212** and spring **228**, the angled surfaces **213** of the braze caps **210***a* and **210***b* engage the radial force member **221** and force the radial force member to push the radial bearing elements **206** to seat within the sockets **204** of the radial bearing ring **202**. The radial force member **221** can be a roller, balls, wedges, or another structure configured to receive force from the braze caps **210***a* and **210***b* and direct at least a portion of the force to the radial bearing elements **206**.

[0055] To secure the radial bearing ring **202** in the clamp assembly **222**, the radial bearing ring **202** is positioned onto the base plate **224**. The radial bearing ring **202** includes sockets **204** configured to receive the radial bearing elements **206**. The braze material **208** is then positioned within sockets **204** on the radial bearing ring **202**. The radial bearing elements **206** are then positioned within the sockets **204** such that the braze material **208** is positioned at least partially between the radial bearing ring **202** and the radial bearing elements **206**. The braze material **208** is positioned between a bottom surface **205** of the radial bearing element **206** and a bottom surface **203** of the socket **204**. A surface **223** of the radial force member **221** is engaged with a top surface **207** of the radial bearing element **206**. Thus, the radial force member **221** is arranged to force the radial bearing element **206** into the socket **204**.

[0056] With the radial bearing ring **202** and radial bearing elements **206** in place, the braze caps **210***a* and **210***b* are secured over the radial bearing ring **202**. The radial force member **221** is positioned between the braze caps **210***a* and **210***b* and the radial bearing elements **206** such that the radial bearing elements **206** are positioned between the radial force member **221** and the radial bearing ring **202**. With the braze caps **210***a* and **210** and radial force member **221** in place, the braze caps **210***a* and **210***b* are secured in position. In the embodiment depicted, the braze caps **210***a* and **210***b* are secured via a bolt **212**, spring **228**, and washer **230**. The bolt **212** is engaged through the spring **228**, washer **230**, clearance holes **226***a* and **226***b*, and radial bearing ring **202**. The bolt **212** is threaded with the base plate **224** at threaded hole **220**.

[0057] The clamp assembly **222** secures the radial bearing elements **206** to the radial bearing ring **202** during the brazing process to facilitate accurate and precise positioning of the radial bearing elements **206** on the radial bearing ring **202**. The bolt **212** and spring **228** exert force on the braze caps **210***a* and **210***b*, which in-turn exert forces via the angled surfaces **213** onto the radial force member **221**. The radial force member **221**, in-turn, exerts force onto the radial bearing elements **206**, pressing the radial bearing elements **206** toward the radial bearing ring **202** to maintain the radial bearing elements **206** in position within the sockets **204** during the brazing process.

[0058] With the radial bearing ring **202** and radial bearing elements **206** secured in place by the clamp assembly **222** at the desired compressive force, the brazing of each of the radial bearing elements **206** can be conducted using the flame ring **214**. Like flame ring **114**, the flame ring **214** includes gas manifold **232**, gas inlet **234** in fluid communication with the gas manifold **232**, nozzle fittings **236** in fluid communication with the gas manifold **232**, and nozzles **238** in fluid communication with the nozzle fittings **236** and configured to direct flames **216** toward the radial bearing ring **202**, radial bearing elements **206**, and braze material **208**. **102**. The flames **216** at least partially melt the braze material **208** to braze the radial bearing elements **206** to the radial bearing ring **202**. As with the thrust bearing brazing, during the application of the flames **216**, one or both of the radial bearing ring **202** and flame ring **214** are rotated.

[0059] Once the brazing of the radial bearing elements **206** is completed, the clamping assembly **222** can be removed by unthreading the bolt **212**, and the radial bearing ring **202** can be removed from the brazing system **2000**. In some embodiments, after the radial bearing elements **206** are brazed to the radial bearing ring **202**, no post-processing is performed on the brazed radial bearing elements **206**.

Brazing Angular Bearings

[0060] With reference to FIGS. **3**A-**3**F, a method and system for securing angular bearing elements to an angular bearing ring is described. Brazing system **3000** is configured to secure an angular bearing ring for brazing of the angular bearing elements thereto. In FIGS. **3**A-**3**F, the brazing system **3000** is depicted brazing angular bearing elements **306** to an angular bearing ring **302**.

[0061] Brazing system **3000** includes a clamp assembly **322** and a flame ring **314**. The clamp assembly **322** is similar to the clamp assembly **122** shown in FIGS. **1**A-**1**G, but is configured to secure a position of the angular bearing ring **302** and angular bearing elements **306** and to apply a compressive force to the angular bearing elements **306** during the brazing process. The flame ring **314** is substantially similar, and may be identical in embodiments, to the flame ring **114** shown in FIGS. **1**A-**1**G. Flame ring **314** is configured to provide flames to heat the angular bearing ring **302**, angular bearing elements **306**, and braze material **308** to achieve the brazing of the angular bearing elements **306** onto the angular bearing ring **302**.

[0062] The clamp assembly **322** includes a chuck **300** having a chuck jaw **318**. A base plate **324** of the clamp assembly **322** is positioned and secured in the chuck jaws **318**. Base plate **324** includes a threaded hole **320**. The clamp assembly **322** is different from the clamp assembly **322** in that the braze cap **310** includes an angled surface **313**. The angled surface **313** may be a beveled surface, chamfered edge, or another transitional surface between the bottom surface **311** and the side edge **315** of the braze cap **310**. The braze cap **310** includes a clearance hole **326**. The angled surface **313** is configured to translate forces of the clamp assembly **322** that are directed in a axial direction into an angled direction toward the angular bearing elements **306**.

[0063] To secure the angular bearing ring **302** in the clamp assembly **322**, the angular bearing ring **302** is positioned onto the base plate **324**. The angular bearing ring **302** includes sockets **304** configured to receive the angular bearing elements **306**. The braze material **308** is then positioned within sockets **304** on the angular bearing ring **302**. The angular bearing elements **306** are then positioned within the sockets **304** such that the braze material **308** is positioned at least partially between the angular bearing ring **302** and the angular bearing elements **306**. The braze material **308** is positioned between a bottom surface **305** of the angular bearing element **306** and a bottom surface **303** of the socket **304**. Angled surface **313** of the braze cap **310** is engaged with a top surface **307** of the angular bearing element **306**. Thus, the angled surface **313** is arranged to force the angular bearing element **306** into the socket **304**.

[0064] With the angular bearing ring **302** and angular bearing elements **306** in place, the braze cap **310** is secured over the angular bearing ring **302**. The angular bearing elements **306** are positioned between the angled surface **313** and the angular bearing ring **302**. With the braze cap **310** in place, the braze cap **310** is secured in position via a bolt **312**, spring **328**, and washer **330**. The bolt **312** is engaged through the spring **328**, washer **330**, clearance hole **326**, and angular bearing ring **302**. The bolt **312** is threaded with the base plate **324** at threaded hole **320**.

[0065] The clamp assembly **322** secures the angular bearing elements **306** to the angular bearing ring **302** during the brazing process to facilitate accurate and precise positioning of the angular bearing elements **306** on the angular bearing ring **302**. The bolt **312** and spring **328** exert force on the braze cap **310**, which in-turn exert forces via the angled surface **313** onto the angular bearing elements **306**, pressing the angular bearing elements **306** toward the angular bearing ring **302** to maintain the angular bearing elements **306** in position within the sockets **304** during the brazing process.

[0066] With the angular bearing ring **302** and angular bearing elements **306** secured in place by the clamp assembly **322** at the desired compressive force, the brazing of each of the angular bearing elements **306** can be conducted using the flame ring **314**. Like flame ring **114**, the flame ring **314** includes gas manifold **332**, gas inlet **334** in fluid communication with the gas manifold **332**, nozzle fittings **336** in fluid communication with the gas manifold **332**, and nozzles **338** in fluid communication with the nozzle fittings **336** and configured to direct flames **316** toward the angular bearing ring **302**, angular bearing elements **306**, and braze material **308**. The flames **316** at least partially melt the braze material **308** to braze the angular bearing elements **306** to the angular bearing ring **302**. As with the thrust bearing brazing, during the application of the flames **316**, one or both of the angular bearing ring **302** and flame ring **314** are rotated.

[0067] Once the brazing of the angular bearing elements **306** is completed, the clamping assembly **322** can be removed by unthreading the bolt **312**, and the angular bearing ring **302** can be removed from the brazing system **3000**. In some embodiments, after the angular bearing elements **306** are brazed to the angular bearing ring **302**, no post-processing is performed on the brazed angular bearing elements **306**.

Burner System

[0068] FIG. **4** is a simplified schematic of a gas/compressed air mixer system for supplying gas to the flame rings disclosed herein. Gas/compressed air mixer system **440** includes an air filter/regulator **442** for supply of air **444** (e.g., from a compressed air source). Valve **446** (e.g., air needle valve) can be used to control the amount of air **444** supplied to the flame ring. System **440** includes a zero gas governor **448** to supply flammable gas **443** (e.g., propane, natural gas) to the flame ring **414**. Valve **450** (e.g., air needle valve) can be used to control the amount of gas **443** supplied to the flame ring **414**. Both the gas **443** and air **444** pass through a mixer **452** (e.g., venturi mixer) for mixing of the gas **443** and air **444**. The air/gas mixture **453** is directed from the mixer **452** to the burners of the flame ring **414**. The gas/compressed air mixer system of FIG. **4** is only exemplary, and the system disclosed herein are not limited to this particular gas and air delivery system. The type of gas, the amount of air, the amount gas, and the proportion of gas and air in the mixture **453** can be varied to vary the flame and heat.

[0069] As the braze material melts, the spring forces the bearing elements into the sockets of the bearing ring. In some embodiments, at least some of the braze material is visible during the brazing process such that a user can visually confirm melting of the braze material. Once the brazing procedure is complete, the clamp assembly can be removed from the bearing housing.

Induction Brazing

[0070] In some embodiments, the heat source is an induction heater instead of a flame ring. For example, in place of the flame rings shown in FIGS. **1**A-**3**E, multi-wrap induction coils can be used to heat the bearing rings, bearing elements, braze material, and/or flux paste. The bearing elements can be positioned within sockets such that a flux paste is positioned between at least a portion of the bearing elements and at least a portion of the bearing ring. With the bearing elements secured within the sockets via a clamp assembly, a multi-wrap induction coil can be used to apply heat to the bearing elements, braze material, flux paste, and bearing ring. The application of heat from multi-wrap induction coil at least partially melts the braze material and flux paste, brazing the bearing elements to the bearing ring. The braze material flows into the space between the bearing elements and bearing ring via capillary action and gravity. The method includes ceasing the application of heat and allowing the braze material, bearing ring, and bearing elements to cool. In some embodiments, the application of induction heat is performed without rotating the bearing housing, with the bearing housing positioned on a non-rotating surface plate. The bearing elements brazed using induction heating can be angular bearings (i.e., not exclusively radial and not exclusively thrust). Prior to application of the heat and the brazing procedure, the braze material is not positioned between the bearing elements and the bearing ring.

Applications

[0071] The brazing methods disclosed herein can be used to secure bearing elements to a variety of different types of bearing assemblies (e.g., sliding bearings) including radial bearings, thrust bearings, combined radial and thrust bearings, angular bearings, and conical bearings. The bearings may be bearings in a downhole drilling tool or motor, or other rotating part. In some embodiments, the bearing element are polycrystalline diamond compacts that include a polycrystalline diamond (PCD) table on a support (e.g., a tungsten carbide support). While the brazing methods disclosed herein are described as being used to braze bearing elements onto rotational bearings, the methods described herein are not limited to this particular application. The brazing methods can be used to apply elements to power transmission surfaces and drivelines.

[0072] Although the present embodiments and advantages have been described in detail, it should be understood that various changes, substitutions and alterations can be made herein without departing from the spirit and scope of the disclosure. Moreover, the scope of the present application is not intended to be limited to the particular embodiments of the process, machine, manufacture, composition of matter, means, methods and steps described in the specification. As one of ordinary skill in the art will readily appreciate from the disclosure, processes, machines, manufacture, compositions of matter, means, methods, or steps, presently existing or later to be developed that perform substantially the same function or achieve substantially the same result as the corresponding embodiments described herein may be utilized according to the present disclosure. Accordingly, the appended claims are intended to include within their scope such processes, machines, manufacture, compositions of matter, means, methods, or steps.

Claims

1. A method of securing bearing elements to a bearing ring, the method comprising: positioning a plurality of bearing elements on a bearing race of a bearing ring; positioning a braze material on the bearing elements, the bearing body, or combinations thereof; clamping the bearing elements and the bearing body together; while clamping, brazing the bearing elements onto the bearing ring, wherein the brazing comprises applying heat from one or more flames of a burner to the bearing elements, the braze material, the bearing body, or combinations thereof, wherein the heat at least partially melts the braze material.

2. The method of claim 1, wherein the brazing includes surrounding the bearing ring with a plurality of flames of the burner that are directed toward the bearing ring.

3. The method of claim 1, wherein, during the brazing, the bearing ring and bearing elements are rotated relative to the burner and flames.

4. The method of claim 3, wherein the burner is maintained static while the bearing ring and bearing elements are rotated.

5. (canceled)

6. The method of claim 1, wherein the brazing is performed under vacuum.

7. The method of claim 1, wherein the burner comprises a generally ring-shaped gas manifold fluidly coupled with a supply of a gas/air mixture, wherein the bearing ring and bearing elements are positioned in an annulus of the ring-shaped gas manifold during the brazing, and wherein a plurality of flame nozzles are fluidly coupled with the ring-shaped gas manifold and positioned to direct the flames from the nozzles toward the bearing ring and bearing elements.

8. (canceled)

9. (canceled)

10. The method of claim 1, wherein the clamping comprises: positioning the bearing ring and the bearing elements on a base plate; positioning a braze cap over the bearing ring and the bearing elements such that the bearing ring and bearing elements are positioned between the braze cap and the base plate; positioning a force applicator on the braze cap; and forcing the braze cap toward to base plate with the force applicator, wherein forcing the braze cap toward the base plate clamps the bearing elements and the bearing ring together.

11. The method of claim 10, wherein the force applicator comprises a spring positioned on the braze cap such that the braze cap is positioned between the spring and the bearing ring and bearing elements, and wherein the forcing comprises compressing the spring.

12. The method of claim 11, wherein compressing the spring comprises threadably coupling a threaded end of a bolt with the base plate, wherein the bolt extends from the threaded end through the bearing ring, the braze cap, and the spring such that the spring is positioned between a bold head of the bolt and the braze cap, wherein threading the bolt towards the braze cap compresses the spring.

13. (canceled)

14. The method of claim 11, further comprising positioning a heat shield around the spring.

15. (canceled)

16. The method of claim 10, wherein: the bearing elements are thrust bearing elements, and wherein a bottom surface of the braze cap is engaged with a top surface of the thrust bearing elements during the forcing; or wherein the bearing elements are angular bearing elements, and wherein an angled surface of the braze cap is engaged with a top surface of the angular bearing elements during the forcing.

17. (canceled)

18. The method of claim 1, wherein the clamping comprises: positioning the bearing ring and the bearing elements on a base plate; positioning a first braze cap over the base plate and within an annulus of the bearing ring, wherein the first braze cap has an angled surface; positioning a radial force member between the first braze cap and the bearing elements; positioning a second braze cap over the first braze cap and within the annulus of the bearing ring, wherein the first and second braze caps are spaced part from one another, wherein the second braze cap has an angled surface, and wherein the radial force member is positioned between the second braze cap and the bearing elements; positioning a force applicator on the second braze cap; and forcing the first braze cap toward to second braze cap, wherein the angled surfaces of the braze caps engage the radial force member and force the radial force member toward the bearing elements, and wherein the radial force member engages the bearing elements and forces the bearing elements towards the bearing ring.

19. (canceled)

21. The method of claim 1, wherein, after the brazing, no lapping, polishing, grinding, or machining is performed on the brazed bearing elements.

22. (canceled)

23. The method of claim 1, comprising monitoring a temperature of the bearing ring, the bearing elements, or combinations thereof during the brazing.

24. The method of claim 1, wherein the bearing elements comprise polycrystalline diamond compacts.

25. The method of claim 1, where the bearing elements are brazed within sockets of the bearing ring.

26. The method of claim 25, wherein, prior to the brazing, the braze material is positioned outside of the sockets.

27. The method of claim 25, wherein, prior to the brazing, the braze material is positioned inside of the sockets between the bearing elements and the bearing ring.

28-42. (canceled)

43. A system for securing and heating bearing elements during brazing of the bearing elements to a bearing ring, the system comprising: a clamp, the clamp including: a base plate, the base plate defining a surface configured to receive a bearing ring with bearing elements; a braze cap, the braze cap having an engagement surface; a force applicator positioned on the braze cap; an actuator positioned to actuate the force applicator to apply a force to the braze cap to force the braze cap toward the base plate; and a flame ring, the flame ring including: a gas manifold, the gas manifold defining a central annulus; and a plurality of nozzles in fluid communication with the gas manifold, wherein the nozzles are positioned to direct flame from the nozzles toward the central annulus.

44. A method of securing bearing elements to a bearing ring, the method comprising: positioning a bearing element on a bearing race of a bearing ring; positioning a braze material on the bearing element, the bearing body, or combinations thereof; clamping the bearing element and the bearing body together; while clamping, brazing the bearing element onto the bearing ring, wherein the brazing comprises applying heat from one or more flames of a burner to the bearing element, the braze material, the bearing body, or combinations thereof, wherein the heat at least partially melts the braze material.